

Tal Ben Ami

Performance Engineer | Deep Learning & Inference Optimization

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Seeking a full-time role in Yokneam | Available 3-4 days/week until graduation in March 2027

Intel Performance Engineer working on firmware latency, instruction-level bottlenecks, and benchmark reliability. Hands-on CUDA kernel optimization and PyTorch projects spanning LLM benchmarking infrastructure and GPU memory-efficient deep learning.

EXPERIENCE

Intel | Performance Engineer, TDX Design Team

Jan 2026 - Present

- Reduced affected SEAMCALL/TDCALL API latencies to 20-50% below Granite Rapids, reversing initial regressions of 20-70% above that baseline on the newer server platform.
- Traced API performance regressions to pipeline bottlenecks in MOVDIR64B/non-temporal memory operations and SEAMCALL/SEAMRET transitions through cross-platform comparisons.
- Optimize C firmware for Intel Trust Domain Extensions (TDX), targeting API latency below 25 microseconds while interrupts are disabled; improve non-temporal memory-operation throughput.
- Refactored benchmark infrastructure to control experimental variables, reduce run-to-run variation, and improve measurement reliability and code robustness.

TECHNICAL SKILLS

Programming & systems: C/C++, Python, x86 assembly, Linux, OpenMP, Git, CMake; firmware performance analysis and microbenchmarking.

GPU & deep learning: CUDA kernel optimization in C/C++ and Python; PyTorch, Hugging Face Transformers, GPU memory profiling; Docker and Kubernetes.

Technical knowledge: Triton kernel fusion; vLLM inference optimization and KV-cache management; NVIDIA GPU networking and RDMA; CPU affinity, cache locality, and read-copy-update (RCU).

SELECTED PROJECTS

LLM Inference Benchmarking Agent & MCP Server [Agent](#) · [MCP](#)

- Built a Python proof of concept that turns workload goals into llm-d benchmark configurations, orchestrates Kubernetes jobs and parameter sweeps, and exposes benchmarking workflows through an MCP server.
- Implemented schema-validated report analysis, latency-target checks, estimated goodput, and Pareto comparisons to support reproducible latency/throughput evaluation. Validated the local kind-cluster workflow.

CoLA + GaLore: Memory-Efficient Vision Transformers [GitHub](#)

- Combined low-rank CoLA layers with GaLore gradient projection in PyTorch; reduced peak training memory from over 20 GB to 6.45 GB in ViT-Huge profiling experiments (approximately 3× lower).
- Compared baseline, CoLA, GaLore, and combined configurations with GPU memory and iteration-time profiling; evaluated training accuracy on ViT-Tiny using CIFAR-10.

DOMINO: AI-Assisted Emergency Response [GitHub](#)

- Won 2nd place among 25 completed projects at Technion CS Hack Doing Good 2026.
- Co-developed a Flutter/Firebase prototype using Hebrew speech transcription and LLM parsing to structure casualty reports, synchronize field/command/hospital dashboards, and generate medical handoff PDFs.

EDUCATION & HONORS

Technion, Israel Institute of Technology

2022 - Mar 2027 (expected)

B.Sc. in Computer Engineering | Average: 86.6/100 · **Dean's List, Spring 2023/24**

Relevant coursework (grades /100): Accelerated Parallel Computing (99); Deep Learning on Computational Accelerators (94); Computational Accelerators and Accelerated Systems (93); Systems Performance Optimization (91); Parallel and Distributed Programming for Data Processing and Machine Learning (90); Computer Architecture (90); Operating Systems (93).